

Online Experiments for Language Scientists

Lecture 8: Priming and overspecification

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Questions about UG assessment 1?

Ask now!

Office hours for OELS questions:

- Tuesday 4th : 11-1pm, just drop in (1.14 DSB)
- Last chance for email questions:
 - 4pm on Monday 3rd November!

Loy & Smith (2021)

Loy, J. E., & Smith, K. (2021). Speakers Align With Their Partner's Overspecification During Interaction. *Cognitive Science*, 45, e13065.



Jia Loy

(now works in industry)

- Design:
 - Director-matcher game: participant with partner
 - Confederate priming experiments
- Research question: Do people copy their partner's tendency to **overspecify**?

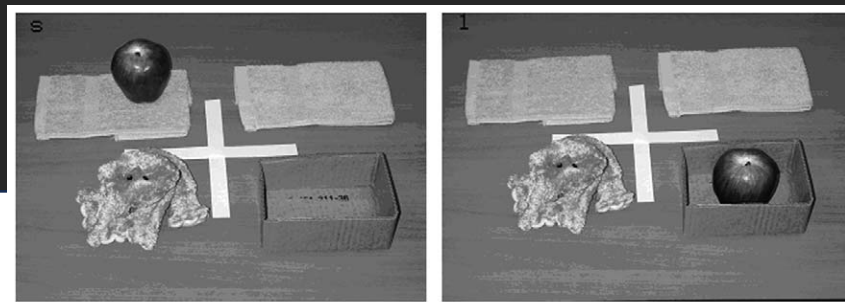
Overspecification

Overspecification: providing more information than what is strictly needed.

e.g., Engelhardt, Bailey & Ferreira (2006) Do speakers and listeners observe the Gricean Maxim of Quantity? *Journal of Memory and Language*, 54, 554-573.

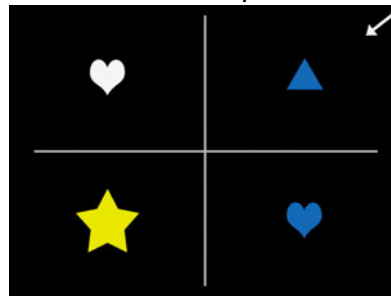
(also Engelhardt & Ferreira 2014)

It's **not** very efficient!

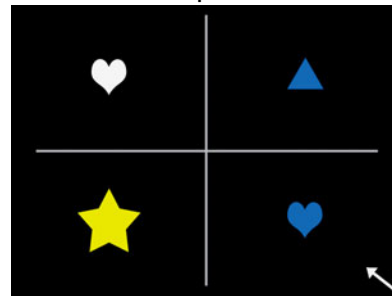


“Put the apple in the box” vs.
“Put the apple that’s on the towel in the box”

A. No modifier required.



B. Modifier required.



“The (blue) triangle” vs.
“The (small) blue heart”

Priming

Priming: people repeat what they have recently heard or produced

Structural priming: people repeat abstract structures they have recently heard or produced

e.g. Bock, J. K. (1986). Syntactic persistence in language production. *Cognitive Psychology*, 18, 355-387.

PRIMING SENTENCES

ACTIVE:

**ONE OF THE FANS
PUNCHED THE
REFEREE.**

PREPOSITIONAL:

**A ROCK STAR SOLD
SOME COCAINE TO AN
UNDERCOVER AGENT.**

PASSIVE:

**THE REFEREE WAS
PUNCHED BY ONE
OF THE FANS.**

DOUBLE OBJECT:

**A ROCK STAR SOLD
AN UNDERCOVER AGENT
SOME COCAINE.**

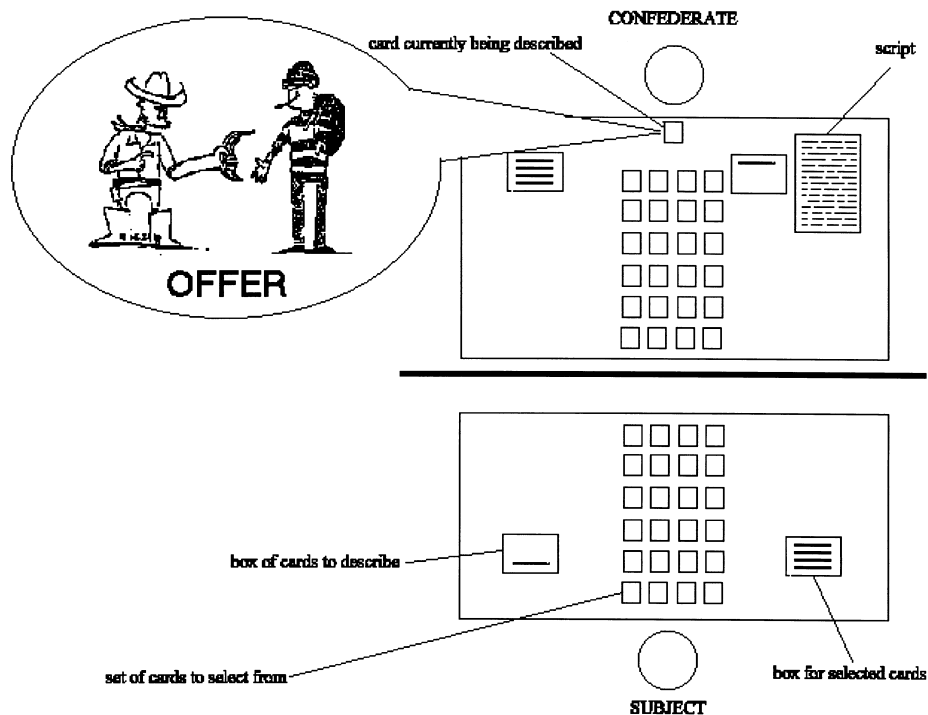
TARGET PICTURES

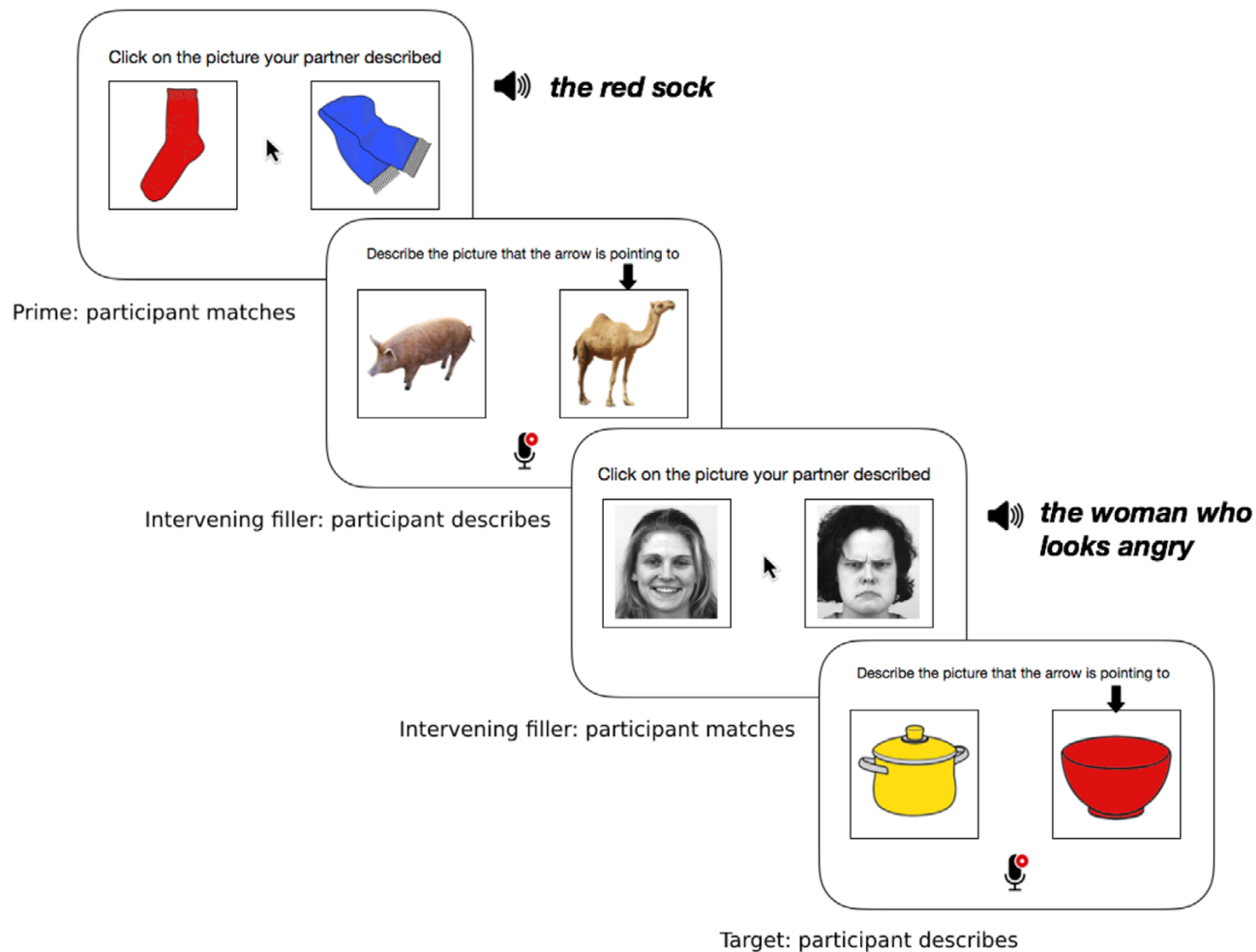


Confederate priming

Confederate: “a person one works with, especially in something secret or illegal; an accomplice”

Branigan, H. P., Pickering, M. J., & Cleland, A. A. (2000). Syntactic coordination in dialogue. *Cognition*, 75, B13-25.





Demo using this week's code

Loy & Smith (2021)

Manipulation: partner's tendency to overspecify

- **Exps 1, 2:** colour, partner either consistently overspecifies (uses colour adjectives) or not (uses bare nouns)
- **Exp 3:** size, partner either consistently overspecifies (uses size adjectives) or not (uses bare nouns)
- **Exp 4, 5:** colour, partner switches behaviour mid-way through experiment

Loy & Smith (2021): details

Exp 1: *lab-based*

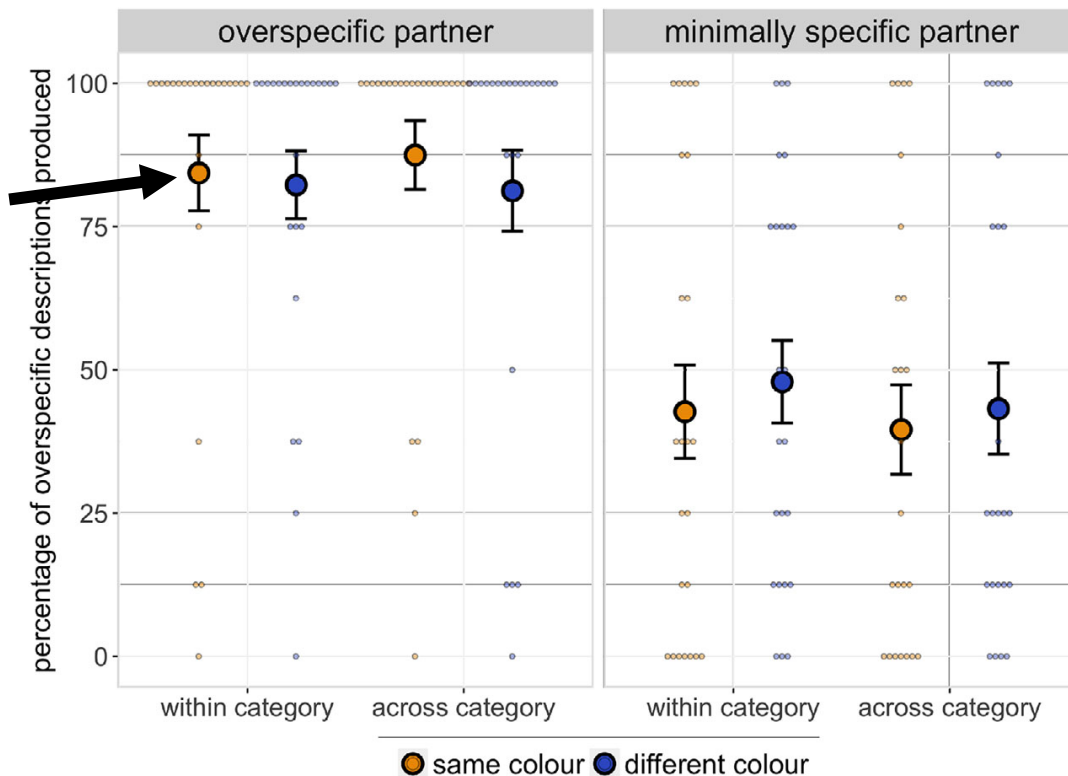
- N=24 per condition after exclusions
- Paid £6

Exps 2-5: *MTurk*

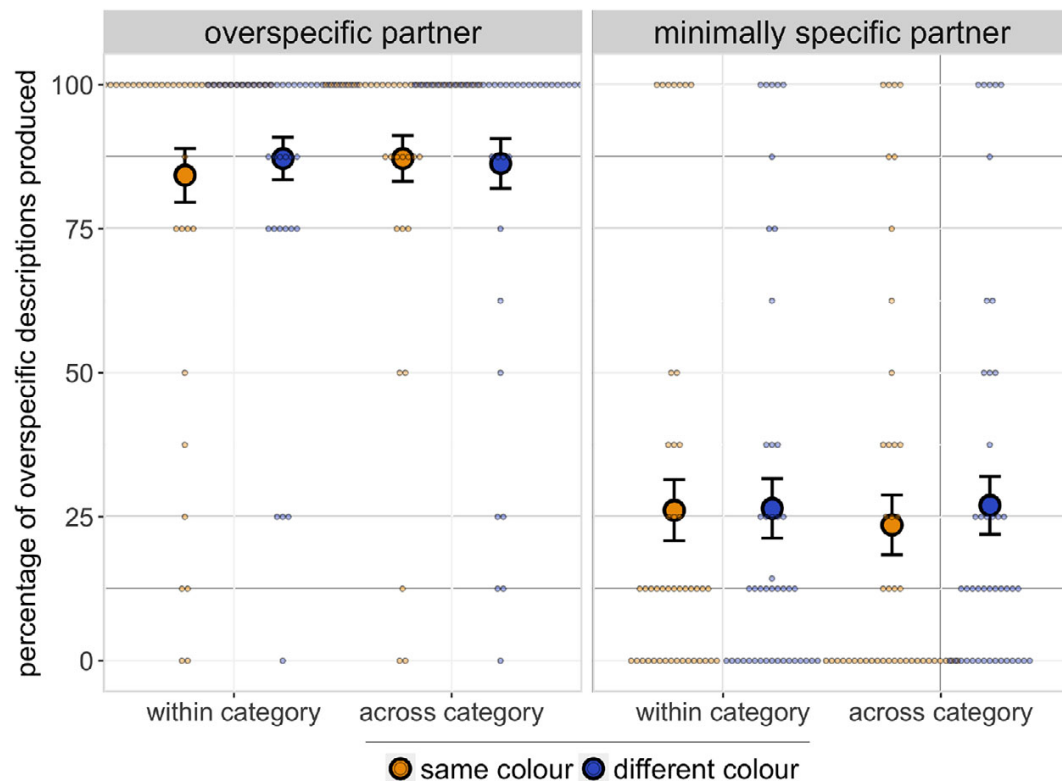
- N≈50 per condition after exclusions
- Paid \$6

Experiment 1: lab, colour

Same colour
Same category
...might expect
more priming

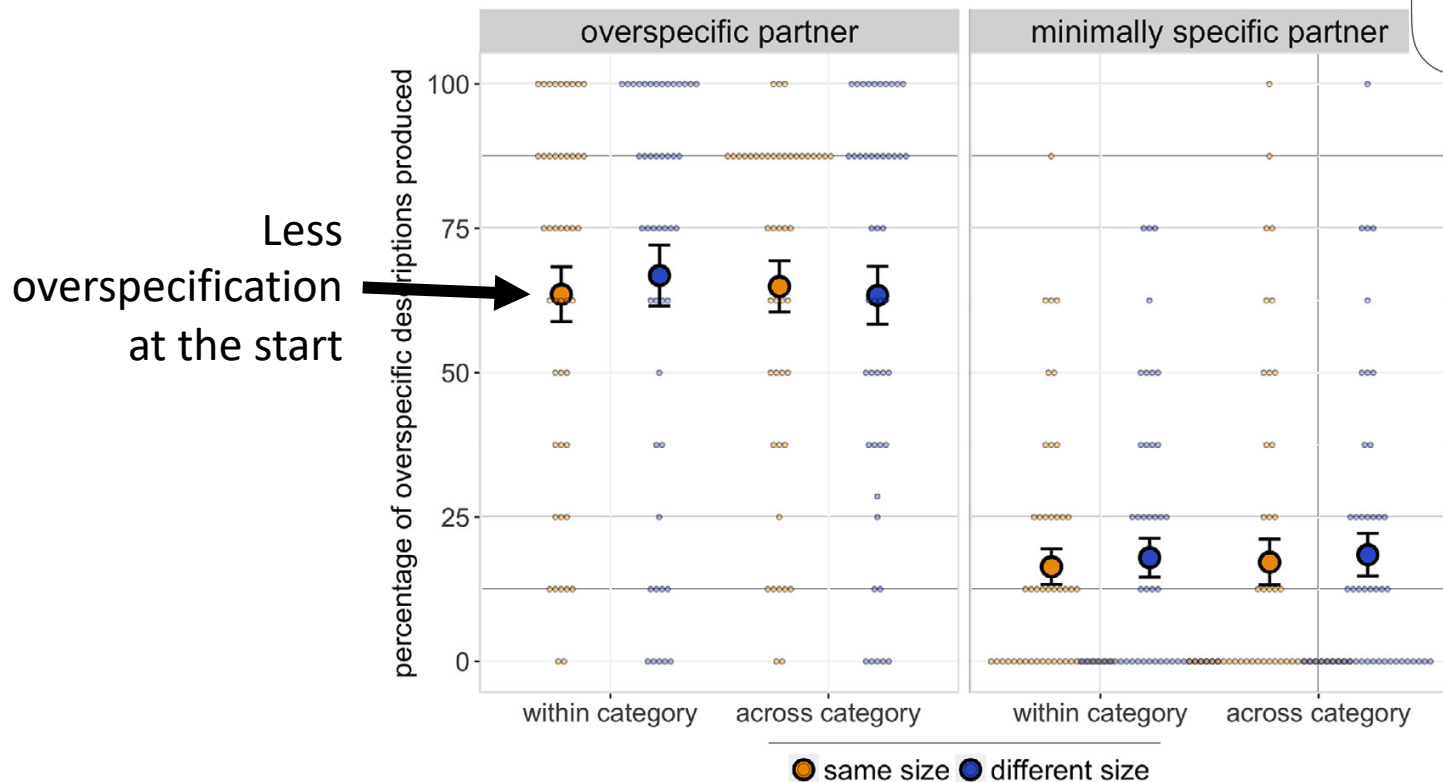
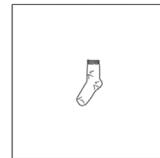
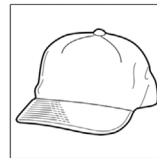


Experiment 2: online, colour

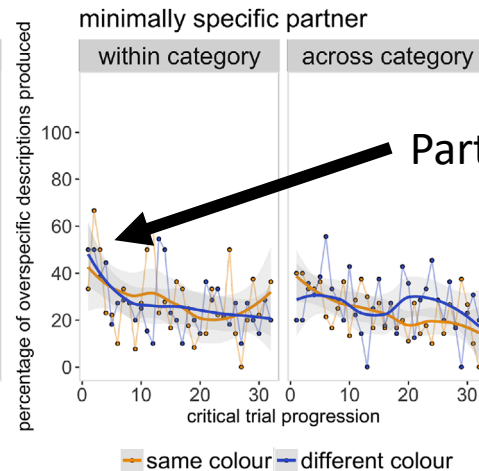
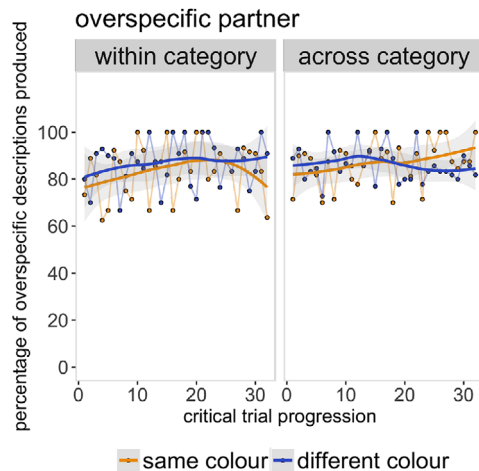


Experiment 3: online, size

Click on the picture your partner described

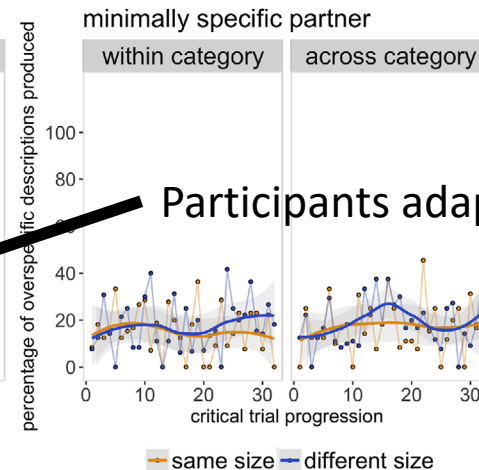
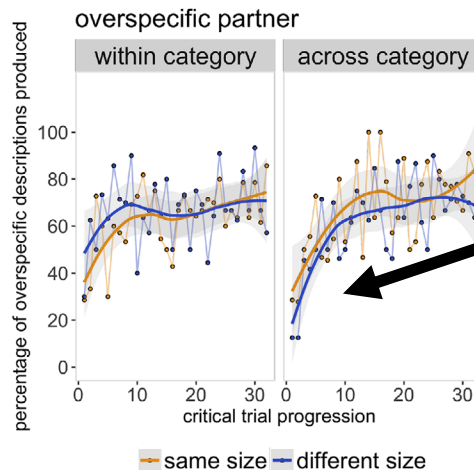


Colour



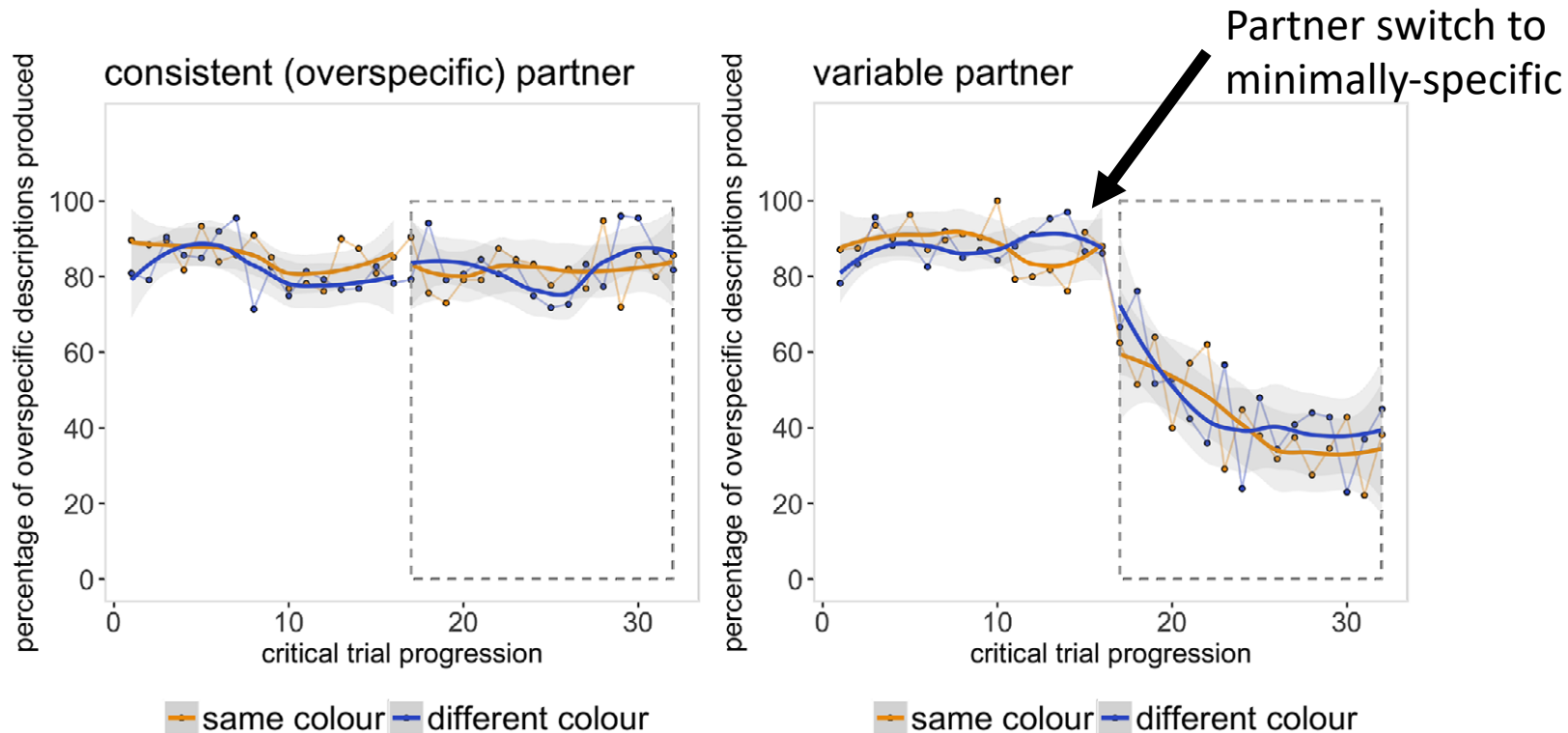
Participants adapt down

Size

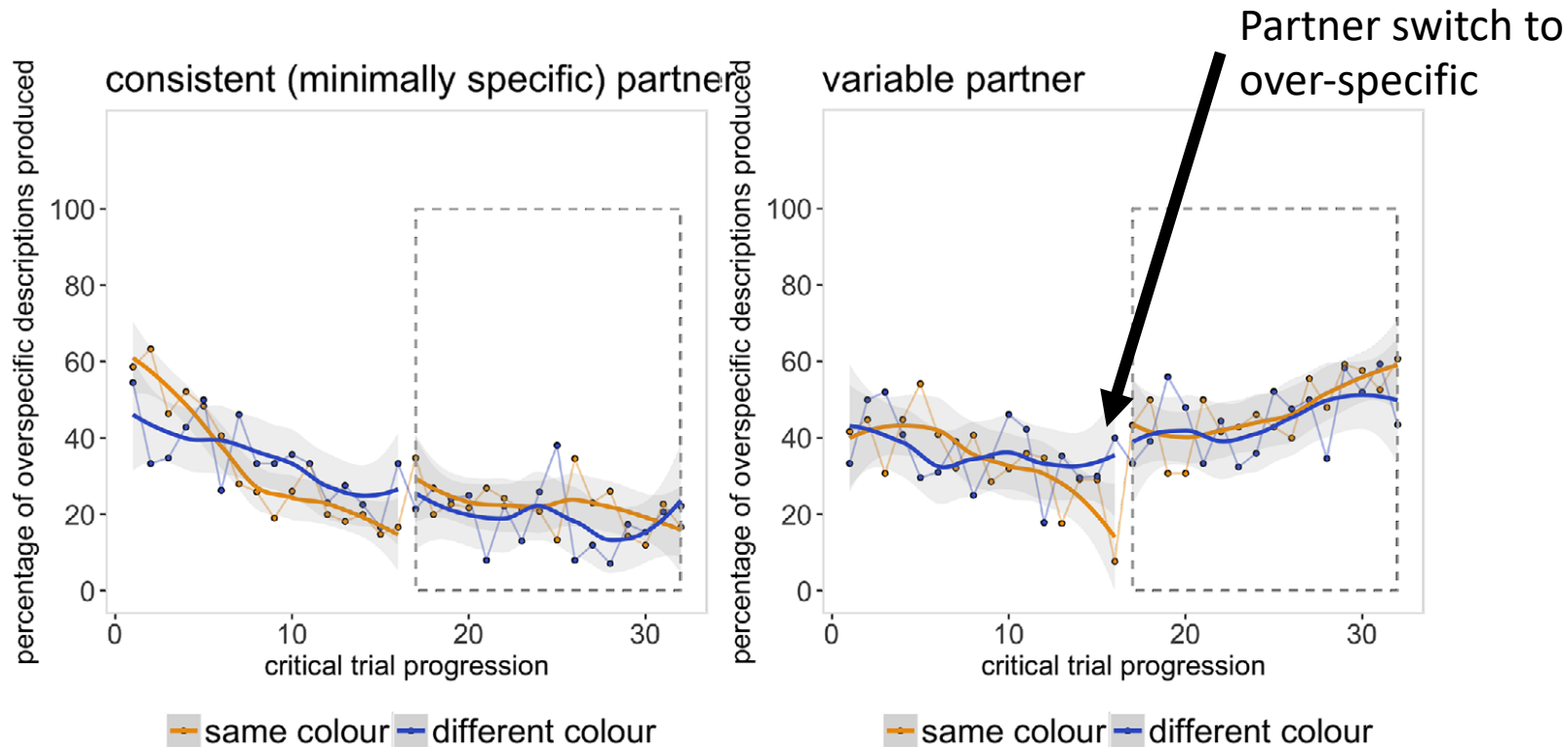


Participants adapt up

Experiment 4: switch to minimally-specific



Experiment 5: switch to over-specific



Loy & Smith's conclusions

1. Overspecifying (or not) can be **primed** during interaction
 - Including if their partner switches behaviour mid-way through the experiment
2. **Social effects** are a *constraint* on people's tendency to behave in an optimally efficient manner in communication

Time for Q&A/discussion on this week's reading

Next up

Lab

- A confederate priming experiment, recording spoken responses

Next week

- Language evolution by iterated learning